Multimodal Sentiment and Emotion Analysis

Multi-Layer Feature Fusion and Multi-Task Learning

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Team TokeNization
Introduction to NLP (S26) — Final Submission

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Paper re-implemented: Cai et al., *Sci. Reports* 2025 (UFEN+MTFN).

Motivation

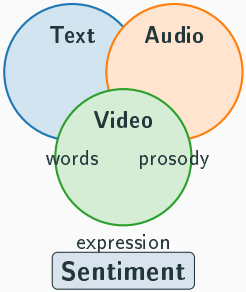
Why Multimodal?

Human communication is multimodal:

- **Text** — what is said
- **Audio** — how it is said
- **Video** — facial reaction

"It was fine."

Text only	→ neutral
+ flat voice	→ negative
+ frown	→ disappointed



Three Core Challenges

1. Heterogeneity

Different feature spaces: BERT 768-d, COVAREP 74-d, FACET 47-d

2. Conflict

Modalities can disagree (sarcasm, irony); simple averaging cancels signal

3. Text Dominance

Text alone $\approx$ 73% on MOSI; model tends to ignore A/V

UFEN+MTFN addresses all three with a two-stage pipeline and five-task supervision.

Related Work & Datasets

Related Work

Category	Representative Methods
Per-modality encoders	BERT, LSTM/GRU, COVAREP, FACET
Tensor fusion	TFN, LMF
Attention fusion	MulT, MFN, MISA
Multi-task MSA	Self-MM, MMIM
ERC on MELD	DialogueRNN, DialogueGCN, MM-DFN, MultiEMO
Our base paper	**UFEN+MTFN** (Cai et al., 2025)

UFEN+MTFN combines MulT-style cross-modal attention with Self-MM-style multi-task learning and adds a Transformer encoder–decoder on top.

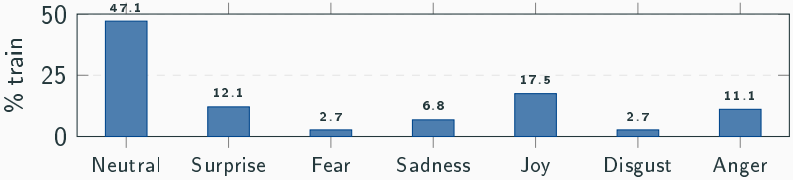
Datasets

CMU-MOSI (Phase 1)

- 2,199 YouTube review clips
- Regression: $[-3, +3]$
- Splits: 1,283 / 214 / 686
- Text: BERT-768 (fine-tuned)
- Audio: COVAREP 74-d
- Visual: FACET 47-d

MELD (Phase 2)

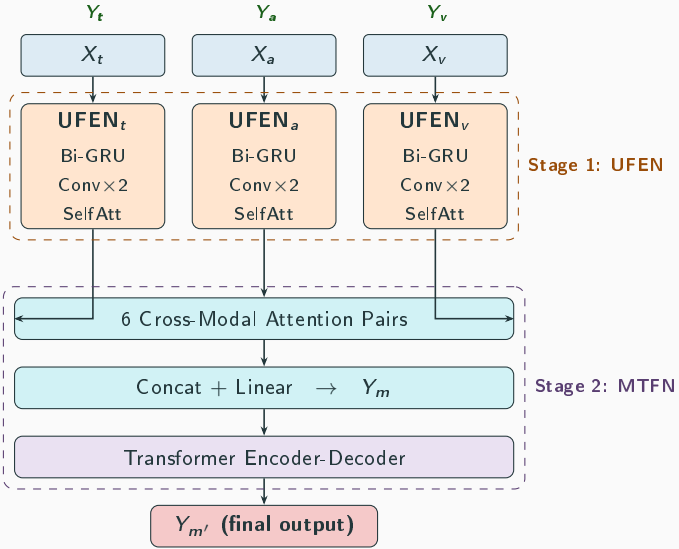
- 13,708 *Friends* utterances
- 7-class classification
- Splits: 9,988 / 1,108 / 2,610
- Strongly imbalanced



MELD label distribution (train split).

Architecture

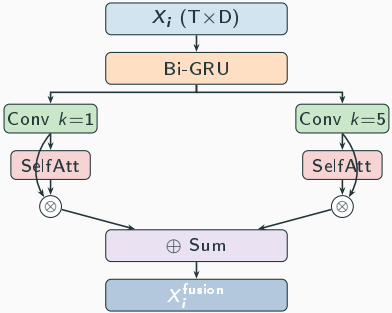
Architecture Overview



Stage 1: UFEN — Unimodal Feature Extraction

One UFEN per modality. Five steps:

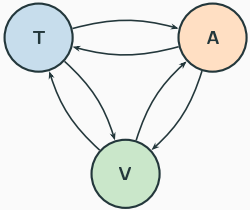
1. **Bi-GRU** captures temporal context in both directions
2. **Parallel Conv1D** branches (kernels $[1, 5]$) extract multi-scale patterns
3. **Self-attention gate**: $X \otimes \mathrm{Att}(X)$ amplifies salient steps
4. **Unpool** + **sum** across branches
5. **Unimodal head** $\rightarrow Y_i$ (used in multi-task loss)



Stage 2: MTFN — Multi-Task Fusion

Six directed cross-modal attention pairs:

$$\mathrm{CA}_{i\to j} = \mathrm{LN}\big(\mathrm{MHAtt}(Q_i, K_j, V_j)\big), \quad (i,j) \in \{t,a,v\}^2,\ i \neq j$$



- Each pair $\to$ mean-pool
- Concat $6 \times d_m \to Y_m$
- Reshape to length-6 seq
- Transformer enc–dec refines
- Final prediction $Y_{m'}$

Multi-Task Loss: 5 Heads, 1 Objective

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_t + \mathcal{L}_a + \mathcal{L}_v}_{\text{unimodal}} + \underbrace{\mathcal{L}_m + \mathcal{L}_{m'}}_{\text{multimodal}}$$

Why 5 losses?

- Without unimodal losses, model ignores A/V
- Text alone reaches ∼73% on MOSI
- Forces each branch to carry signal

Task-specific variants

- Phase 1 (MOSI): MSE, 5 regression heads
- Phase 2 (MELD): Weighted cross-entropy + label smoothing, 5 classification heads

Test-time output is $Y_{m'}$ only; the other four heads exist only during training.

Phase 1: CMU-MOSI

Phase 1: CMU-MOSI — Best Configuration (P3-2)

Search scope: 25+ experiments in 3 sub-phases

- **1a**: Single-param sweeps
- **1b**: Combinations
- **1c**: Regularisation

Key unlocks

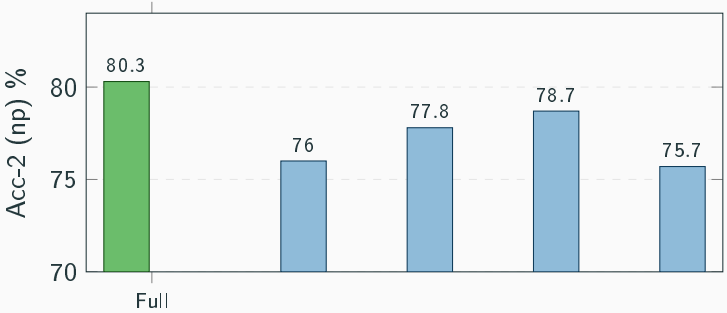
- $d_{\text{ff}} = 128$ (not $2 \times d_m$)
- Kernels $[1, 5]$
- Cosine LR + early stop = 15
- BERT warmup (5 epochs linear)
- Weight decay was catastrophic

Metric	Ours (P3-2)	Paper
MAE ↓	**0.812**	0.728
Corr ↑	**0.745**	0.792
Acc-7 ↑	**43.0**	46.7
Acc-2 nn ↑	**78.6**	85.2
Acc-2 np ↑	**80.3**	86.6
F1 np ↑	**80.4**	86.7

Dev MAE reached 0.7434 — within 0.015 of paper's target.

∼110.8M params (BERT dominates).

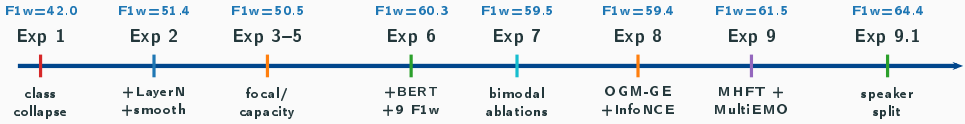
Phase 1 Ablation — Every Component Matters



- **Cross-modal attention** −4.3 points → biggest component
- **Unimodal losses** −2.5 points → stop text collapse
- **Encoder–decoder** −1.6 points → refinement helps
- **Single conv** −4.6 points → multi-scale is essential

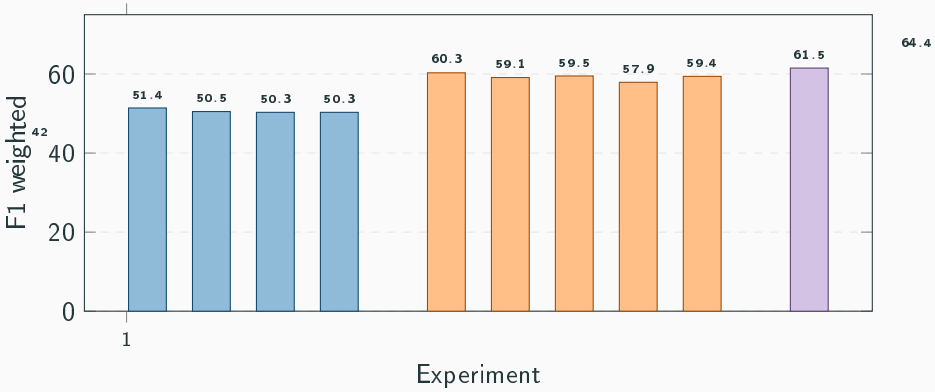
Phase 2: MELD

Phase 2 Experiment Trail — Timeline



Three major jumps: Exp 2 (feature norm), Exp 6 (BERT), Exp 9.1 (MHFT + clean split).

Phase 2 Results — Weighted F1 Progression



Exp 1: collapse **2–5**: GloVe plateau **6–8**: BERT (111M params) **9**: MHFT **9.1**: **best**

Exp 1–2: Diagnosing Class Collapse

Exp 1 failure: only Neutral & Anger predicted

- LR 5e-3 too high (oscillation)
- GloVe L2 norm ≈ 5 vs. audio 0.85, video 0.72 $\rightarrow$ **7× text dominance**
- Trivial minimum: majority class only

Exp 2 fix: 3 targeted changes

- Per-modality LayerNorm
- LR $\rightarrow 10^{-3}$
- Label smoothing 0.1

Result

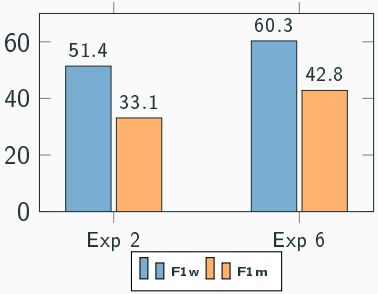
- F1w: 42.0 $\rightarrow$ 51.4 (**+9.4**)
- F1m: 19.6 $\rightarrow$ 33.1 (**+13.5**)
- All 7 classes get predictions

Exp 6: BERT Breaks the GloVe Ceiling

Why GloVe plateaued: context-free embeddings cannot disambiguate

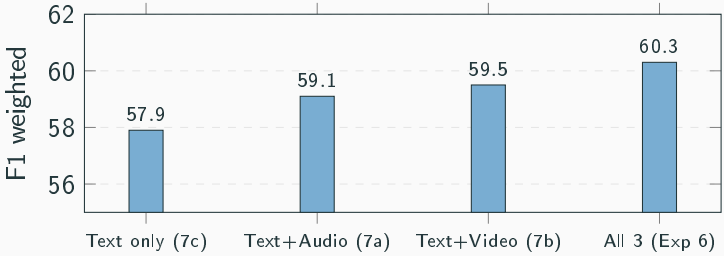
- *"Yeah, right"* — sarcasm or agreement?
- *"I can't believe it"* — joy, anger, surprise?

Exp 6: `bert-base-uncased`, fine-tuned end-to-end, differentiated LR (BERT 2e-5, rest 1e-3), warmup + cosine.



+8.9 F1w, +9.7 F1m
(every class improved)

Exp 7: Bimodal Ablations — Who Contributes What?



- **Text alone** = 57.9 F1w — very close to full 3-modal (60.3)
- Audio & video together add only **+2.4 F1w** over text
- Audio shines on **Sadness** (prosody); video on **Neutral** (face grounding)
- *Friends* is scripted — emotion is encoded in word choice

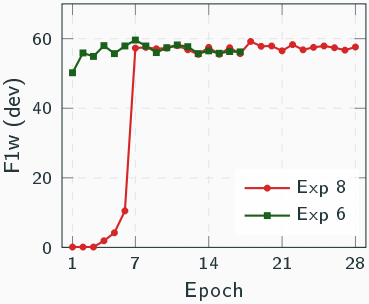
Exp 8: OGM-GE + InfoNCE + Curriculum — A Wash

Three research-backed interventions stacked:

1. **OGM-GE** (Peng et al., CVPR 2022): rescale weak-branch gradients
2. **InfoNCE** (CLIP-style): contrastive alignment across modality pairs
3. **Curriculum**: freeze text → unfreeze → full

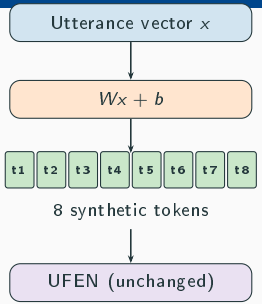
Result: F1w 60.3 → 59.4 (−0.9)

Diagnosis: Phase A (A+V only) gave **2% accuracy** over 3 epochs — the weak modalities cannot bootstrap alone on MELD.



Phases A & B are wasted compute.

Exp 9: MHFT Front-End — The Breakthrough



F1w: 60.3 → 61.5

Params: 111M → 3.58M (30× shrink)

Problem: MultiEMO features are utterance-level (one vector per clip) — UFEN expects a time sequence.

Solution: **MHFT**. Project 1 vector → K=8 tokens:

$$\mathrm{MHFT}(x) = \mathrm{reshape}(Wx + b) + P \in \mathbb{R}^{8 \times d_m}$$

UFEN and MTFN run **unchanged** on the synthetic sequence.

Features: RoBERTa-768 (T) + openSMILE-512 (A) + DenseNet-1000 (V)

Exp 9.1: Closing the Dev→Test Gap

Problem: Exp 9 had dev F1w = 71.7 but test = 61.5 — a **10-point** generalisation gap.

Root cause: Random split of `trainVid` leaked *Friends* speakers into dev. Model memorised speaker-specific patterns.

Three fixes:

1. Speaker-disjoint split

Hold out whole dialogue groups by dominant speaker

Gap: 10.2 → 5.7

2. Capped class weights

Cap Fear/Disgust weights at 1.5 (was 2.24 / 2.41)

False positives: 84 → 43

3. Top-5 ensemble

Average softmax of top-5 dev checkpoints

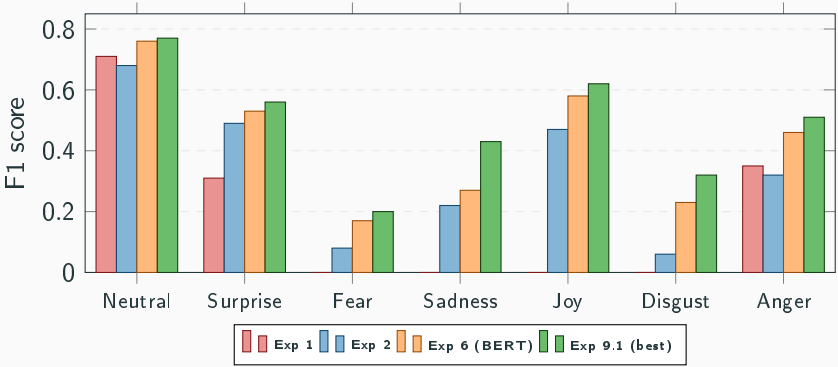
+0.9 F1m free

	Exp 9	**Exp 9.1**	**Δ**
Test F1w	61.5	**64.4**	+2.9
Test F1m	45.7	**49.0**	+3.3
Train size	9,920	6,464	−3,456

Less training data, better generalisation.

Analysis

Per-Class F1 — Exp 9.1 vs. Baselines



Exp 9.1 wins or ties on **5/7 classes**. Largest gains: Sadness (+0.08), Disgust (+0.07), Anger (+0.04).

What Did *Not* Help — and Why

Attempted	Δ F1w	Root cause
Focal loss ($\gamma = 2$)	-0.9	Over-penalises easy Neutral/Joy, hurts mid-frequency classes
Video projection 2048→256	-1.1	BiGRU implicitly selects channels end-to-end; fixed projection discards
Doubled capacity ($d_m = 256$)	-1.1	7.1M params on 10K samples → instability (best ep = 3)
Lower LR (5×10^{-4})	-1.1	Slower convergence, no better final
Scheduled unfreezing	-0.9	A/V cannot bootstrap alone (2% accuracy in Phase A)

Lesson: on small datasets with one dominant modality, small surgical fixes beat elaborate interventions.

Error Analysis — Exp 9.1 Confusion Patterns

Three persistent patterns:

1. **Neutral absorbs errors** — ∼7–10% of every other class pulled into Neutral
2. **Anger/Surprise/Joy triangle** — shared high-arousal prosody
3. **Fear & Disgust** still hard (50 & 68 test samples)

	Neu	Sur	Fear	Sad	Joy	Dis	Ang
Neutral	**912**	65	43	60	84	28	64
Surprise	24	**177**	4	4	27	10	35
Fear	9	4	**17**	6	5	1	8
Sadness	40	14	26	**83**	12	6	27
Joy	59	40	8	6	**253**	10	26
Disgust	14	5	4	2	0	**28**	15
Anger	44	41	17	13	29	22	**179**

Diagonal = correct predictions; green = strong, orange = weak.

Wrapping Up

Key Takeaways

1. **UFEN+MTFN architecture is sound** — every component is ablation-justified on MOSI.
2. **BERT is the dominant lever on MELD**: context-free embeddings plateau at ∼51 F1w; fine-tuned BERT jumps it to 60.3.
3. **MHFT is the surprise winner**: 3.58M params beats 111M BERT on every MELD metric. Pre-extracted features + a lightweight tokenization bridge is a very favourable trade.
4. **Evaluation integrity matters as much as architecture**: swapping a random dev split for speaker-disjoint closed a 4-point gap with *less* training data.
5. **Negative results are informative**: focal loss, doubled capacity, and curriculum unfreezing all failed — each teaches something about the dataset.

Limitations & Future Work

Limitations

- Small datasets (MOSI: 1.3K; MELD: 10K train)
- Single-seed reporting (no std)
- Paper hyperparameters under-specified
- Wave2Vec2-32 & ResNet-101 are not SOTA for affect
- No inter-utterance context

Future Work

- Multi-seed mean $\pm$ std
- Stronger A/V extractors: WavLM, Whisper, AffectNet
- Dialogue context (DialogueRNN-style)
- MHFT on time-aligned features
- Extend to MOSEI, IEMOCAP, CH-SIMS

Ethical note: MELD is drawn from *Friends*, a US sitcom with a narrow demographic and scripted emotion. Deployed systems need re-validation on diverse populations and spontaneous speech.

Thank you!

Questions?

Code: `github.com/Gautam-Gandhi/multimodal-sentiment-analysis`

Team TokeNization: **Manas**, **Soham**, **Parth**, **Gautam**